Most Popular Azure Services

Complete Summary Guide

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Introduction

Microsoft Azure offers over 200 cloud services across various categories. This guide covers the most popular and widely adopted services that form the foundation of modern cloud deployments. Understanding these core services is essential for anyone working with Azure infrastructure and applications.

Compute Services

Azure's compute services provide the processing power for applications and workloads.

Azure Virtual Machines

What it is: Infrastructure-as-a-Service (IaaS) offering that provides on-demand, scalable computing resources.

Key Features:

- Wide variety of VM sizes and configurations
- Support for Windows and Linux operating systems
- Pay-as-you-go or reserved instance pricing
- Integration with Azure networking and storage

Use Cases:

- Migrating on-premises servers to the cloud
- Development and testing environments
- Running legacy applications
- High-performance computing workloads

Azure App Service

What it is: Platform-as-a-Service (PaaS) for building, deploying, and scaling web applications.

Key Features:

- Support for multiple programming languages (.NET, Java, PHP, Python, Node.js)
- Built-in CI/CD integration

- Auto-scaling capabilities
- Custom domain and SSL certificate support

Use Cases:

- Web applications and APIs
- Mobile app backends
- RESTful services
- Microservices architecture

Azure Functions

What it is: Serverless compute service for event-driven applications.

Key Features:

- Pay only for execution time
- Automatic scaling based on demand
- Multiple trigger types (HTTP, timer, queue, etc.)
- Support for various programming languages

Use Cases:

- Data processing and transformation
- IoT data processing
- Real-time file processing
- API backends for mobile applications

Azure Container Instances (ACI)

What it is: Simple container hosting service without the need for orchestration.

Key Features:

- Fast container startup times
- Per-second billing
- Support for both Linux and Windows containers
- Integration with Azure Virtual Networks

Use Cases:

- Simple containerized applications
- Batch processing jobs

- Development and testing
- Event-driven processing

Azure Kubernetes Service (AKS)

What it is: Managed Kubernetes service for container orchestration.

Key Features:

- Managed Kubernetes control plane
- Integrated monitoring and logging
- Auto-scaling and auto-upgrades
- Integration with Azure DevOps and GitHub Actions

Use Cases:

- Microservices applications
- Container orchestration at scale
- DevOps and CI/CD pipelines
- Hybrid and multi-cloud deployments

Storage Services

Azure storage services provide scalable, secure, and durable data storage solutions.

Azure Blob Storage

What it is: Object storage service for unstructured data.

Key Features:

- Massively scalable object storage
- Multiple access tiers (Hot, Cool, Archive)
- Integration with Azure CDN
- Advanced security and compliance features

Use Cases:

- Backup and disaster recovery
- Data lake storage for analytics
- Content distribution
- Application data storage

Azure Files

What it is: Managed file shares accessible via SMB and NFS protocols.

Key Features:

- Fully managed file shares in the cloud
- Cross-platform compatibility
- Integration with on-premises Active Directory
- Snapshot capabilities for backup

Use Cases:

- Lift-and-shift applications requiring file shares
- Shared storage for applications
- Development environments
- Configuration files and logs

Azure Disk Storage

What it is: High-performance block storage for Azure Virtual Machines.

Key Features:

- Multiple disk types (Premium SSD, Standard SSD, Standard HDD)
- Encryption at rest and in transit
- Snapshot and backup capabilities
- High IOPS and throughput options

Use Cases:

- Virtual machine storage
- Database storage
- High-performance applications
- Enterprise workloads

Azure Data Lake Storage

What it is: Scalable storage service optimized for big data analytics workloads.

Key Features:

- Hierarchical namespace for efficient data organization
- Integration with Apache Hadoop ecosystem

- Advanced security and access control
- Optimized for analytics workloads

Use Cases:

- Big data analytics
- Data warehousing
- Machine learning data preparation
- IoT data storage

Database Services

Azure provides managed database services for various data models and use cases.

Azure SQL Database

What it is: Fully managed relational database service based on SQL Server.

Key Features:

- Automatic backups and point-in-time restore
- Built-in high availability
- Automatic tuning and performance optimization
- Advanced security features

Use Cases:

- Modern cloud applications
- Data warehousing
- Line-of-business applications
- E-commerce platforms

Azure Cosmos DB

What it is: Globally distributed, multi-model NoSQL database service.

Key Features:

- Multiple data models (document, key-value, graph, column-family)
- Global distribution with multi-region writes
- Multiple consistency levels
- Automatic and instant scalability

Use Cases:

- Globally distributed applications
- Real-time analytics
- IoT and mobile applications
- Gaming and social media platforms

Azure Database for MySQL/PostgreSQL

What it is: Fully managed database services for popular open-source databases.

Key Features:

- Built-in high availability
- Automatic backups and monitoring
- Security features and compliance
- Flexible scaling options

Use Cases:

- Web applications
- Content management systems
- E-commerce applications
- Data analytics workloads

Azure Cache for Redis

What it is: Fully managed in-memory caching service.

Key Features:

- High-performance in-memory caching
- Multiple pricing tiers and configurations
- Built-in clustering and replication
- Integration with Azure applications

Use Cases:

- Application performance improvement
- Session state management
- Real-time analytics
- Gaming leaderboards

Networking Services

Azure networking services provide secure connectivity and traffic management.

Azure Virtual Network (VNet)

What it is: Private network infrastructure in Azure.

Key Features:

- Isolated network environment
- Subnet segmentation
- Network security groups
- VPN and ExpressRoute connectivity

Use Cases:

- Secure application hosting
- Hybrid cloud connectivity
- Network isolation and segmentation
- Multi-tier application architectures

Azure Load Balancer

What it is: Layer 4 load balancing service for high availability.

Key Features:

- High availability and scalability
- Internal and external load balancing
- Health probes and automatic failover
- Support for TCP and UDP traffic

Use Cases:

- Distributing traffic across multiple VMs
- High availability architectures
- Scaling applications horizontally
- Regional load distribution

Azure Application Gateway

What it is: Layer 7 load balancer with web application firewall capabilities.

Key Features:

- SSL termination and end-to-end SSL

- Web Application Firewall (WAF)
- URL-based routing
- Cookie-based session affinity

Use Cases:

- Web application load balancing
- SSL offloading
- Web application security
- Microservices routing

Azure VPN Gateway

What it is: Service for establishing secure connections between Azure and on-premises networks.

Key Features:

- Site-to-site and point-to-site VPN connections
- Multiple VPN protocols supported
- High availability options
- Integration with Azure Virtual Network

Use Cases:

- Hybrid cloud connectivity
- Remote user access
- Branch office connectivity
- Secure communication channels

Identity & Security Services

Azure identity and security services protect resources and manage access.

Azure Active Directory (Entra ID)

What it is: Cloud-based identity and access management service.

Key Features:

- Single sign-on (SSO) capabilities
- Multi-factor authentication (MFA)
- Conditional access policies
- Integration with thousands of SaaS applications

Use Cases:

- User identity management
- Application access control
- Device management
- B2B and B2C scenarios

Azure Key Vault

What it is: Secure storage service for secrets, keys, and certificates.

Key Features:

- Hardware security module (HSM) backed options
- Access policies and RBAC integration
- Audit logging and monitoring
- Integration with Azure services

Use Cases:

- Application secrets management
- Certificate lifecycle management
- Encryption key storage
- Compliance requirements

Azure Security Center (Microsoft Defender for Cloud)

What it is: Unified security management and threat protection platform.

Key Features:

- Security posture assessment
- Threat detection and response
- Compliance dashboard
- Security recommendations

Use Cases:

- Cloud security monitoring
- Compliance management
- Threat detection and response
- Security policy enforcement

AI & Machine Learning Services

Azure AI services enable intelligent applications and data insights.

Azure Cognitive Services

What it is: Collection of pre-built AI APIs for common scenarios.

Categories:

- **Vision:** Computer vision, face recognition, OCR
- **Speech:** Speech-to-text, text-to-speech, translation
- **Language:** Text analytics, translation, QnA Maker
- **Decision:** Content moderation, anomaly detection

Use Cases:

- Adding AI capabilities to applications
- Document processing and analysis
- Customer service automation
- Content moderation

Azure Machine Learning

What it is: End-to-end machine learning lifecycle platform.

Key Features:

- Automated machine learning (AutoML)
- MLOps capabilities
- Model deployment and management
- Integration with popular ML frameworks

Use Cases:

- Building custom ML models
- Model deployment and monitoring
- Data science collaboration
- Predictive analytics

Analytics & Big Data Services

Azure analytics services enable data-driven insights and decision making.

Azure Synapse Analytics

What it is: Enterprise data warehouse and big data analytics platform.

Key Features:

- Massive parallel processing (MPP)
- Integration with Azure Data Lake
- Built-in Apache Spark
- Real-time and batch analytics

Use Cases:

- Enterprise data warehousing
- Big data analytics
- Real-time stream processing
- Business intelligence

Azure Data Factory

What it is: Data integration service for ETL/ELT workflows.

Key Features:

- Code-free data integration
- Support for 90+ data connectors
- Data flow transformations
- Monitoring and alerting

Use Cases:

- Data migration and integration
- ETL/ELT processes
- Data pipeline orchestration
- Hybrid data integration

Power BI

What it is: Business intelligence and data visualization platform.

Key Features:

- Interactive dashboards and reports
- Natural language queries

- Mobile accessibility
- Integration with Azure services

Use Cases:

- Business reporting and analytics
- Self-service BI
- Data visualization
- Executive dashboards

DevOps & Management Services

Azure DevOps and management services streamline development and operations.

Azure DevOps

What it is: Complete DevOps toolchain for software development.

Components:

- **Azure Repos:** Git repositories
- **Azure Pipelines:** CI/CD pipelines
- **Azure Boards:** Work item tracking
- **Azure Artifacts:** Package management
- **Azure Test Plans:** Testing tools

Use Cases:

- Software development lifecycle management
- Continuous integration and deployment
- Project management and tracking
- Code collaboration

Azure Monitor

What it is: Comprehensive monitoring and observability platform.

Key Features:

- Application and infrastructure monitoring
- Log analytics and queries
- Alerting and notifications
- Performance insights

Use Cases:

- Application performance monitoring
- Infrastructure monitoring
- Log analysis and troubleshooting
- Proactive alerting

Integration Services

Azure integration services connect applications and data across hybrid environments.

Azure Logic Apps

What it is: Workflow automation and system integration service.

Key Features:

- Visual workflow designer
- 400+ pre-built connectors
- Serverless execution model
- Integration with Azure and third-party services

Use Cases:

- Business process automation
- System integration
- Data synchronization
- Event-driven workflows

Azure Service Bus

What it is: Enterprise messaging service for decoupled applications.

Key Features:

- Reliable message delivery
- Support for queues and topics
- Dead letter queues
- Message sessions and transactions

Use Cases:

- Microservices communication

- System decoupling
- Message buffering
- Enterprise integration patterns

Remember: These popular Azure services form the foundation of most cloud solutions. Start with understanding the core services in each category, then expand based on your specific requirements and use cases.

This document is part of the Azure training materials by Dan Mill Training.